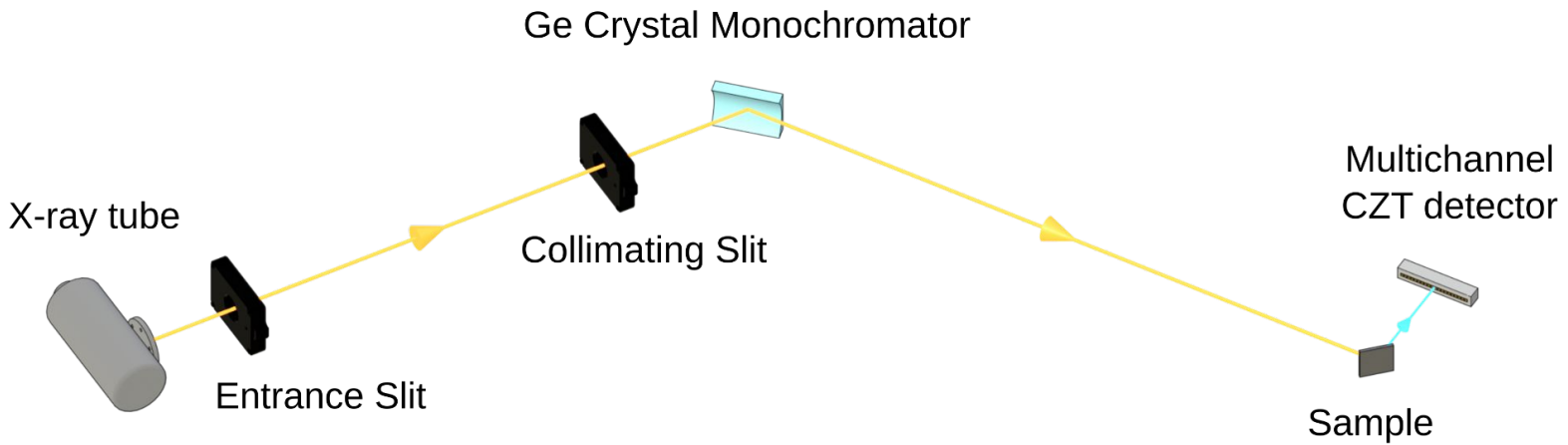
PRISM

Precision Room-temperature Instrumentation for high-energy X-ray absorption Spectroscopy Measurements

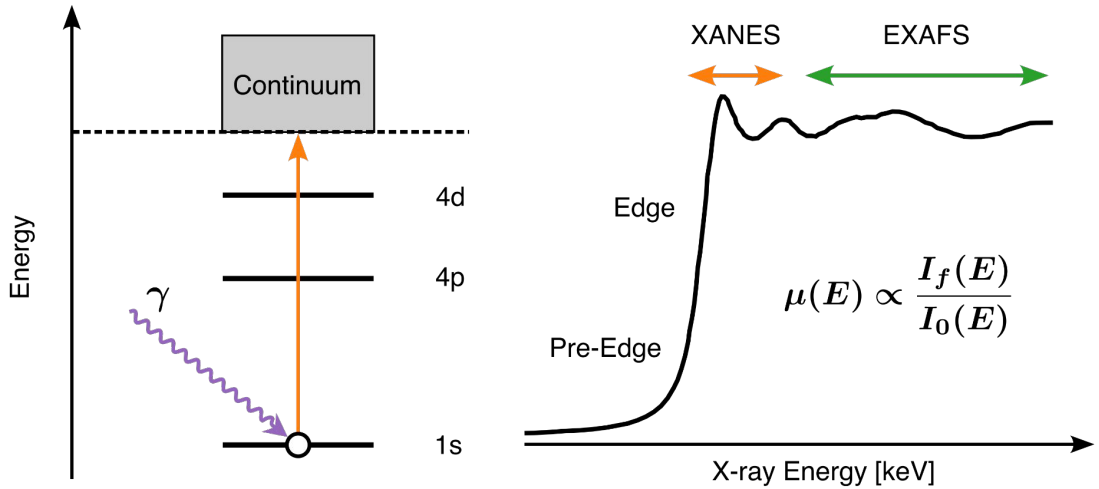
Responsabile di Progetto: Dr. Simone Manti, INFN - LNF
Referente di Progetto: Dr.ssa Catalina Curceanu, INFN - LNF

How can we perform laboratory XAS above 20 keV?



X-ray Absorption Spectroscopy (XAS): chemical state and local structure

XAS probes element-specific **Absorption Edges** to reveal **Chemical State** and the **Local Atomic Environment**.



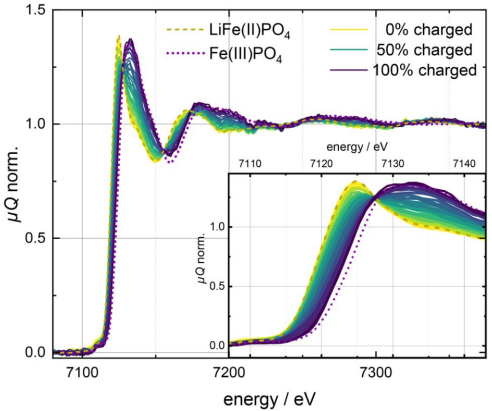
1. **Element selectivity**
   Core-level edges identify the absorbing element
2. **EXAFS**
   Neighbor distances, coordination and disorder
3. **Pre-edge**
   Electronic states and local symmetry
4. **XANES**
   Oxidation state and coordination geometry

State-of-the-Art XAS under operating conditions

Element-specific insight into **Electronic State** and **Local Atomic Structure**.

Battery cycling

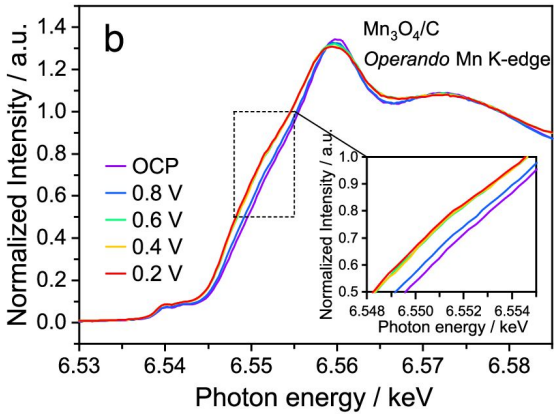
Lab XANES tracks $LiFePO_4$ conversion
Fe K-edge, 15 min per spectrum



Praetz et al., JAAS 40, 2447–2461 (2025), Fig. 2

Working catalysts

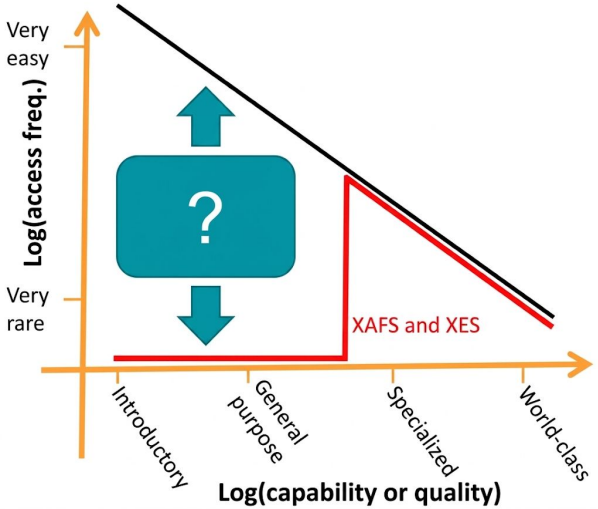
Operando XANES probes the Mn state
under fuel-cell operating conditions



Li et al., Nature Communications (2025), Fig. 3b

Why bring XAS into the laboratory?

Synchrotrons and **FEL** remain the reference for XAS performance.
Laboratory instruments with X-ray tubes offer routine access and flexible measurement times.



J. Seidler, et al. (2024)

- **Routine screening** and **repeated measurements**
- Long **In-Situ / Operando** campaigns
- Sample and protocol optimization for beamtime

Commercial laboratory XAS already exists!

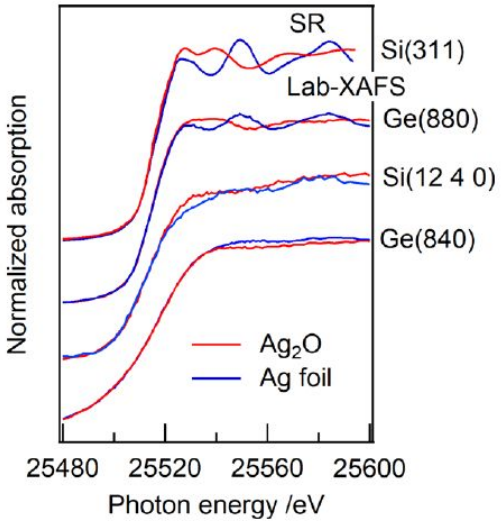


Sigray
≈ 1.5 M€



Laboratory XAS above 20 keV: opportunities and challenges

- K-edges of Mo, Pd, Ag and Sn are relevant to catalysis and energy materials.
- Closely spaced L-edges can restrict the accessible EXAFS range.



Feasibility demonstrated in transmission

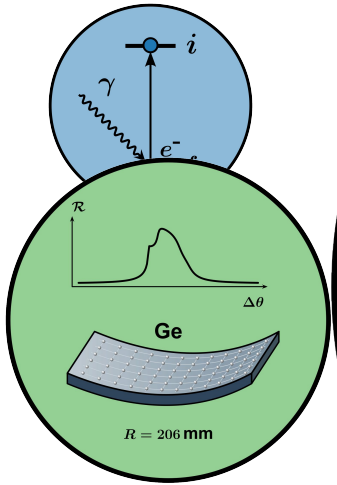
- Ge(880): ≈12 h for Ag K-edge XANES
- Ag K-edge EXAFS impractical at this count rate

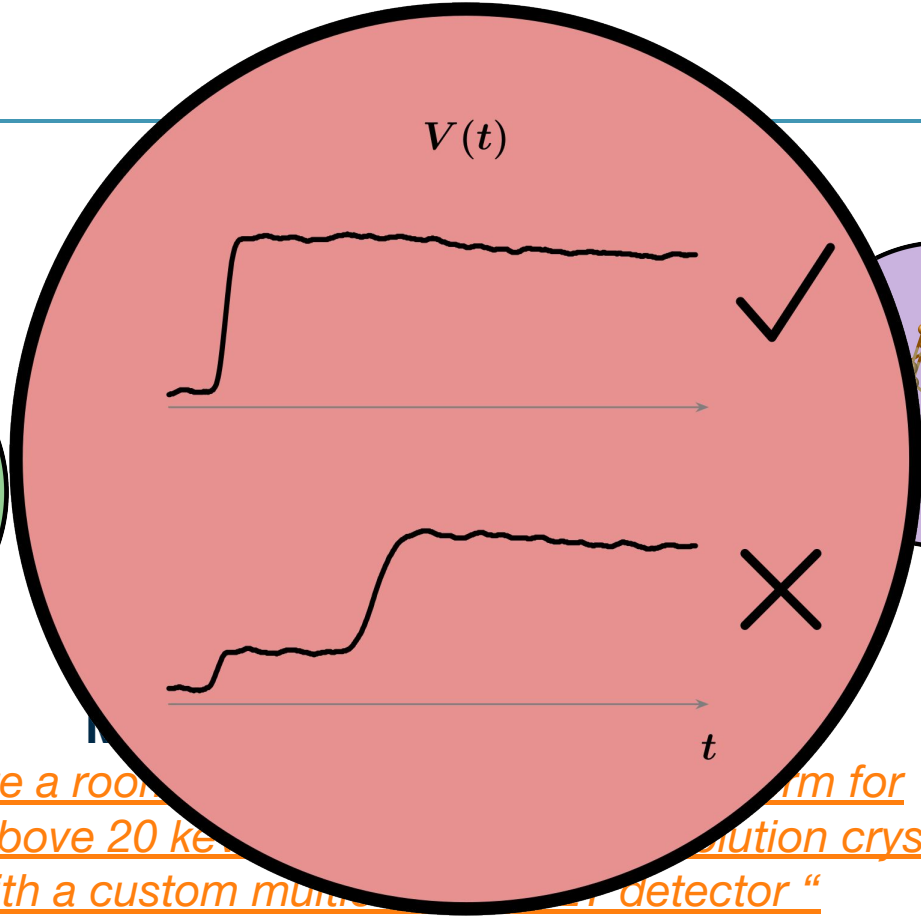
Spectral purity is also critical

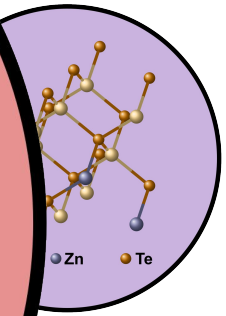
- Al filters suppress lower-order reflections and characteristic Ge K lines.

Challenge: quantitative fluorescence XAS with efficient detection and sufficient SNR!

Yamamoto. "Selection of Monochromator Crystal for Pb L- and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." *X-Ray Spectrometry* (2026).





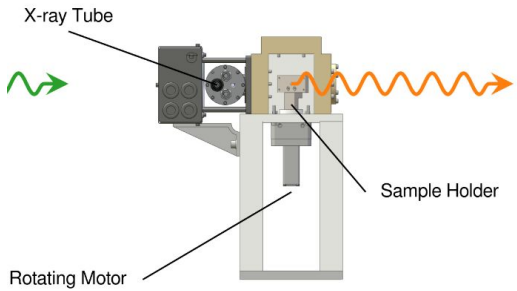


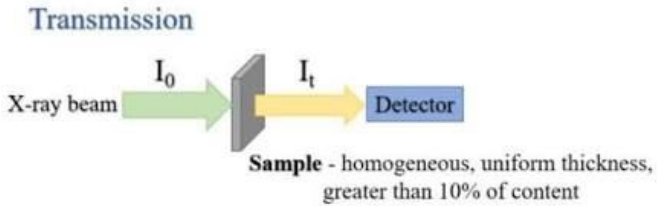
“to develop and validate a roo m for fluorescence-mode XAS above 20 ke olution crystal monochromator with a custom multi detector “

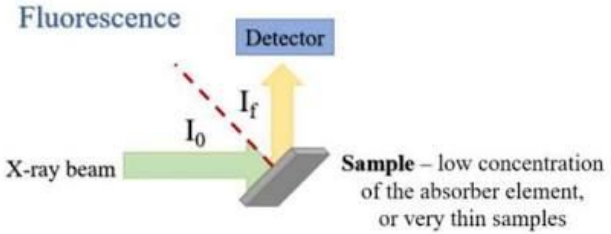
Technique Transmission vs Fluorescence

Transmission fast but limited for thin sample

Fluorescence for any sample but angular scan



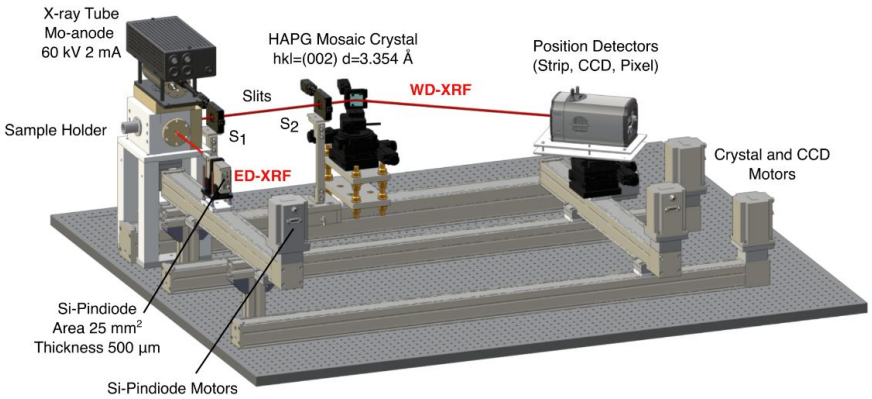




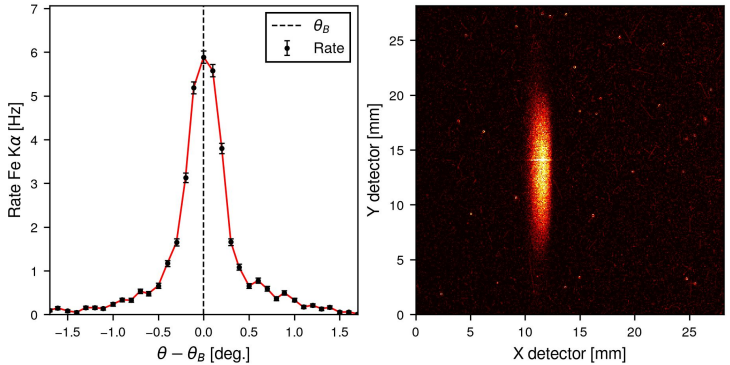
Manti, Simone, and Alessandro Scordo. "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies." *Engineering Innovations* 19 (2026): 23-29.

VOXES capabilities

VOXES

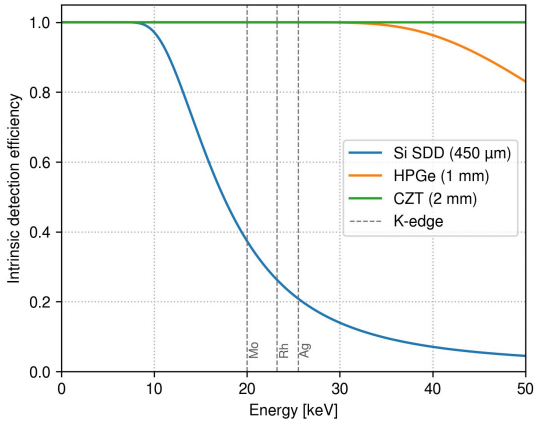


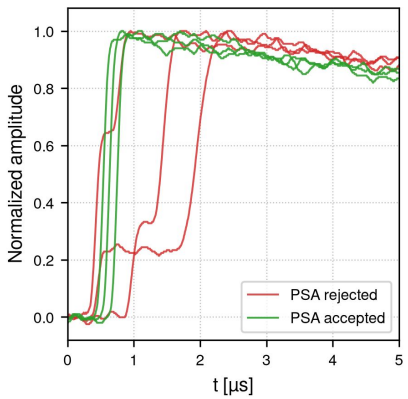


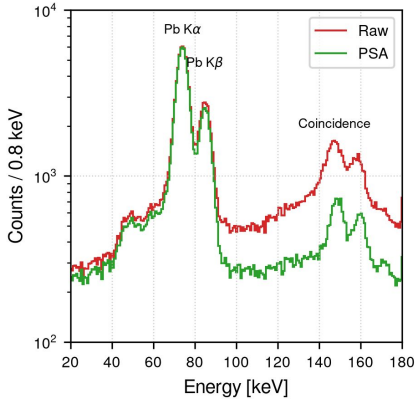


Manti, Simone, et al. "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations." *Condensed Matter* 9.1 (2024): 19.

Custom CZT detector to detect fluorescence above 20 keV





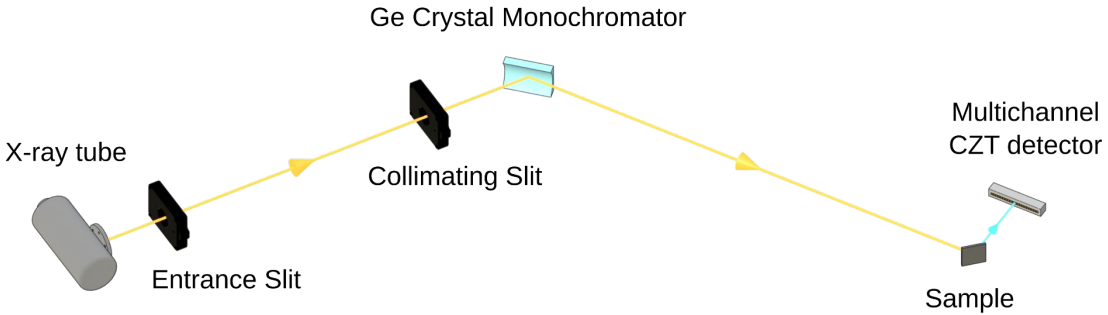


Manti, S. et al. Pulse-Shape Discrimination with Machine Learning for CZT Detectors at DAΦNE Beam Test Facility. *Condens. Matter* **2026**, *11*, 38

Manti, S., et al. "Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment." *Journal of Instrumentation* 21.01 (2026): P01036.

Goal

Measuring the Ag K-edge



Organization

WORK PACKAGE (WP)	PLANNED ACTIVITY (A)	2027				2028			
		1-3	4-6	7-9	10-12	13-15	16-18	19-21	22-24
WP1 - Requirements, simulations and design freeze	A1.1 - Requirements definition								
	A1.2 - SHADOW4 ray-tracing optimization								
	A1.3 - GEANT4 fluorescence simulations								
	A1.4 - Final geometry selection			MS1					
WP2 - Room-temperature multichannel CZT spectroscopy for hard X-ray fluorescence XAS	A2.1 - CZT detector and front-end design								
	A2.2 - Multichannel DAQ and signal-processing development								
	A2.3 - CZT calibration and performance characterization								
	A2.4 - Detector validation and operating-point selection					MS2			
WP3 - Instrument integration and commissioning	A3.1 - Procurement of source, optics and motion stages								
	A3.2 - Mechanical design and spectrometer assembly								
	A3.3 - Hardware, software and control-system integration								
	A3.4 - Instrument commissioning and verification						MS3		
WP4 - Ag reference samples and realistic sample demonstration	A4.1 - Ag foil calibration and reproducibility tests								
	A4.2 - XANES validation on Ag reference compounds								
	A4.3 - Fluorescence XAS demonstration on a realistic sample								
	A4.4 - Performance characterization and operating limits								MS4

Synargies

CSN3: KAONNIS e VIP
CSN5: VOXES (past CSN5)
EuPRAXIA: AQUA and EuAPS

Palermo (CZT) COST ENRICH

Due2Lab, Rigaku e QD

Impact

d



Cost

Year	Category	Item	Purpose/WP	Cost [€]
2027	Equipment	CZT Detector and Front-End	Fluorescence Detection - WP2	€30.000
2027	Equipment	Digitizer and DAQ	Signal Acquisition - WP2	€23.000
2027	Equipment	Motorized Stages and Controllers	Automated Positioning - WP3	€10.000
2027	Consumables	Ag Reference Targets	Calibration Tests - WP2/WP4	€2.000
2027	Services	Custom Mechanical Supports	Instrument Integration - WP3	€2.000
2027	Computing	Data Cloud CPU and Storage	End-to-End Simulations - WP1	€2.000
Total 2027				**€69.000**
2028	Equipment	Cooled X-ray Tube	Excitation Source - WP3	€10.000
2028	Equipment	X-ray Tube Power Supply	Source Operation - WP3	€7.000
2028	Travel	Oral Conference Travel	Results Dissemination - WP4	€4.000
2028	Equipment	Bent Crystal Optics	Energy Selection - WP3	€20.000
2028	Consumables	Additional Reference Targets	Multi-Edge Validation - WP4	€2.000
Total 2028				**€43.000**
Total				**€112.000**

TABLE I. Detailed financial request for the two-year PRISM project, broken down by year, cost category, item, and related Work Package.

Risks

Year	Category	Item	Purpose/WP	Cost [€]
2027	Equipment	CZT Detector and Front-End	Fluorescence Detection - WP2	€30.000
2027	Equipment	Digitizer and DAQ	Signal Acquisition - WP2	€23.000
2027	Equipment	Motorized Stages and Controllers	Automated Positioning - WP3	€10.000
2027	Consumables	Ag Reference Targets	Calibration Tests - WP2/WP4	€2.000
2027	Services	Custom Mechanical Supports	Instrument Integration - WP3	€2.000
2027	Computing	Data Cloud CPU and Storage	End-to-End Simulations - WP1	€2.000
Total 2027				**€69.000**
2028	Equipment	Cooled X-ray Tube	Excitation Source - WP3	€10.000
2028	Equipment	X-ray Tube Power Supply	Source Operation - WP3	€7.000
2028	Travel	Oral Conference Travel	Results Dissemination - WP4	€4.000
2028	Equipment	Bent Crystal Optics	Energy Selection - WP3	€20.000
2028	Consumables	Additional Reference Targets	Multi-Edge Validation - WP4	€2.000
Total 2028				**€43.000**
Total				**€112.000**

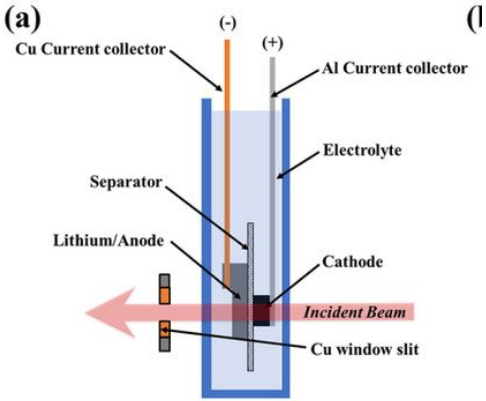
Risk Priority Number

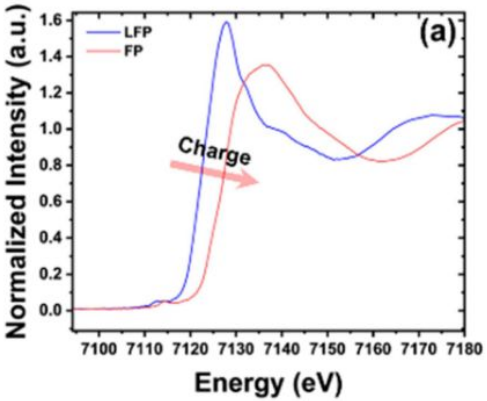
Conclusion

Title

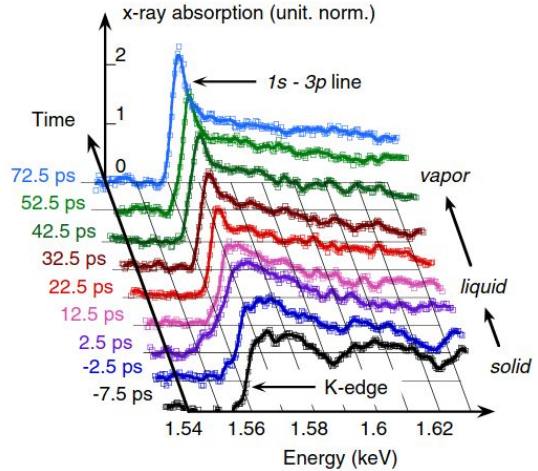
State-of-the-Art XAS: tracking materials during operation

- Element-specific sensitivity to **Oxidation State** and **Local Atomic Structure** enables studies of materials under operating conditions.
- **Battery Cycling:** Operando XAS tracks changes in the Fe oxidation state during charging.
- **Phase Transitions:** Ultrafast XANES follows the evolution from solid to liquid and vapor.





Krishnan et al. *ACS* 16.13 (2024)

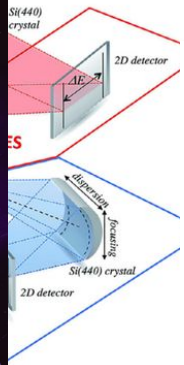


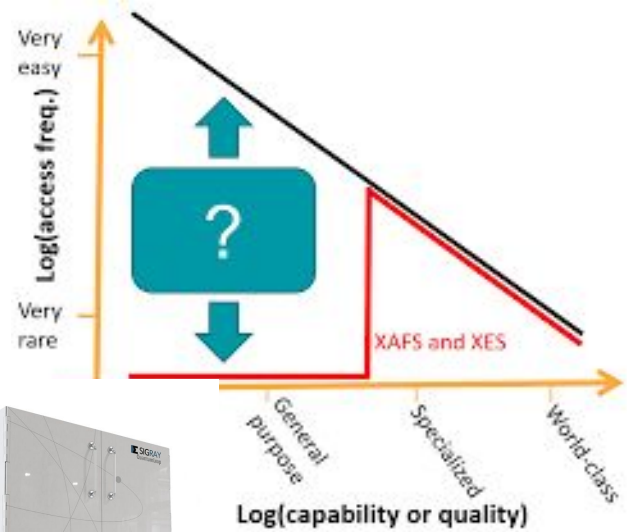
Dorchies, F., et al. *PRL* 107.24 (2011)




easyXAFS300



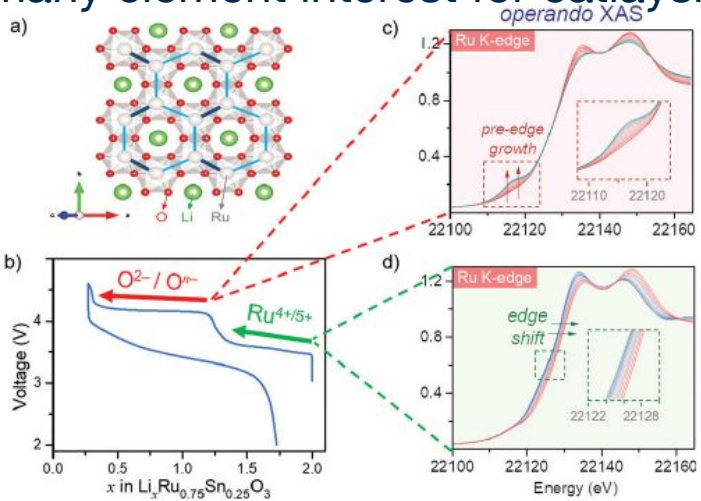

Si(440) crystal
2D detector
ΔE
dispersive
focusing
Si(440) crystal
2D detector

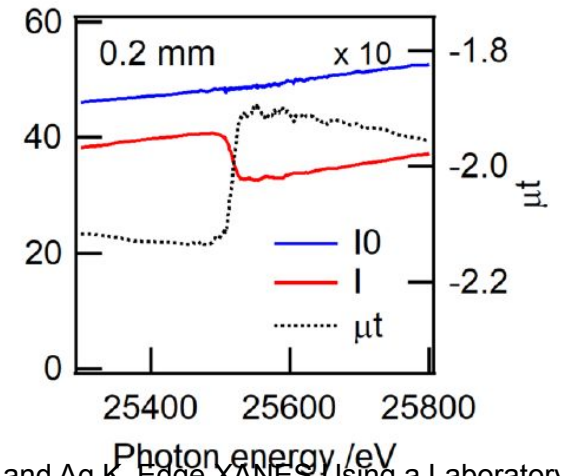

Very easy
Log(access freq.)
Very rare
?
XAFS and XES
General purpose
Specialized
World-class
Log(capability or quality)



Laboratory XAS above 20 keV: opportunities and challenges

- Current limitations are performing XAS for edge above 20 keV, where many element interest for catlaysi (Mo,Pd,Ag,Sn)





Yamamoto, Takashi. "Selection of Monochromator Crystal for Pb L-, Br K-, and Ag K-Edge XANES Using a Laboratory-Based Spectrometer." *X-Ray Spectrometry* (2026).